



SO-DIMM and DIMM are different

on the motherboard as the notch on the connecting sides are positioned differently. Also, if you have two modules with same DDR rating but different frequencies, the motherboard will make them both operate at the frequency of the lower-frequency module. Hence, if you plan to upgrade, get the one with matching frequencies.

Memory capacity is the third most important (and often neglected) part of the spec-sheet. Each motherboard can address only a limited amount of memory in each of its slots. In addition, both Motherboard and CPU have a limitation of how much memory it can address in total. It is futile to install more memory than the lower supported size between the two. E.g. If the CPU supports 24 GB of memory and the Motherboard supports up to 32 GB on 4 available slots, then you will not be able to use a RAM module with size more than 8 GB or more than 24 GB in total. Do not worry, almost every processor and motherboard now support as much as 64 GB of RAM. So you need not worry about this pointer any more.

Latency is the fourth important point we want to mention but it is one of the finest and probably the most neglected part of the compatibility matrix. While two RAM modules may have the same DDR rating and frequency, latencies might make them incompatible. If you were to put two RAM modules in a single channel with same DDR and frequency ratings but different latencies, they might not work. The funny part is: the motherboard may not even show an error. It can simply reject the RAM module; all you will just notice is the total available memory being less than what is physically installed. You might end up thinking that your motherboard has gone rogue and needs replacement while that might not be the case. We have seen this happening and thus recommend that you pay attention to those numbers.

If you are well-versed with overclocking, it goes without saying, the module of your choice must be overclockable! Of course, as always, read the fine print before attempting overclocking. **d**